DEEP LOGIC · AI/ML

NeuroPulse AI

Real-Time Human Error & Fatigue Predictor

> *Predicting fatigue before mistakes happen.*



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The Problem Statement

Stress or Fatigue -The Silent Performance Killer

Fatigue and cognitive overload silently erode focus, increase errors, and degrade performance — often without any visible warning.

Aviation & Transport

Operator alertness is a matter of safety

Healthcare

Professional fatigue risks patient outcomes

Developers & Students

Prolonged sessions spike cognitive decline

Corporate Workforce

Burnout drives attrition and reduced output

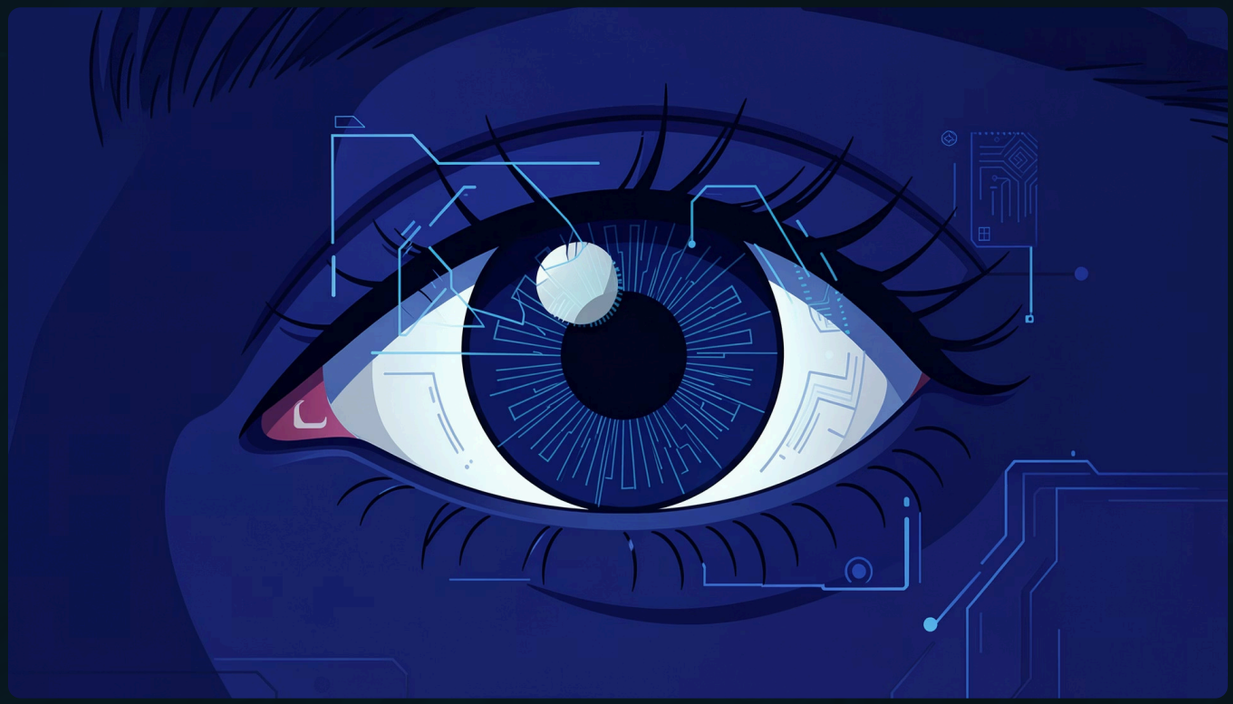
⚠ Most systems react **after errors occur** — when fatigue has already caused damage. The world needs an AI that detects both physical and cognitive fatigue in real time.



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Solution
NeuroPulse AI: Dual Intelligence Engine

A multi-modal fatigue detection platform that fuses two independent AI models into one unified platform.



Vision AI — Webcam Based

- Eye closure detection
- Yawning & facial drowsiness cues
- Continuous alertness scoring

Behavioral AI — Interaction Based

- Typing inconsistency & error correction spikes
- Mouse hesitation & path anomalies
- Session fatigue trend analysis

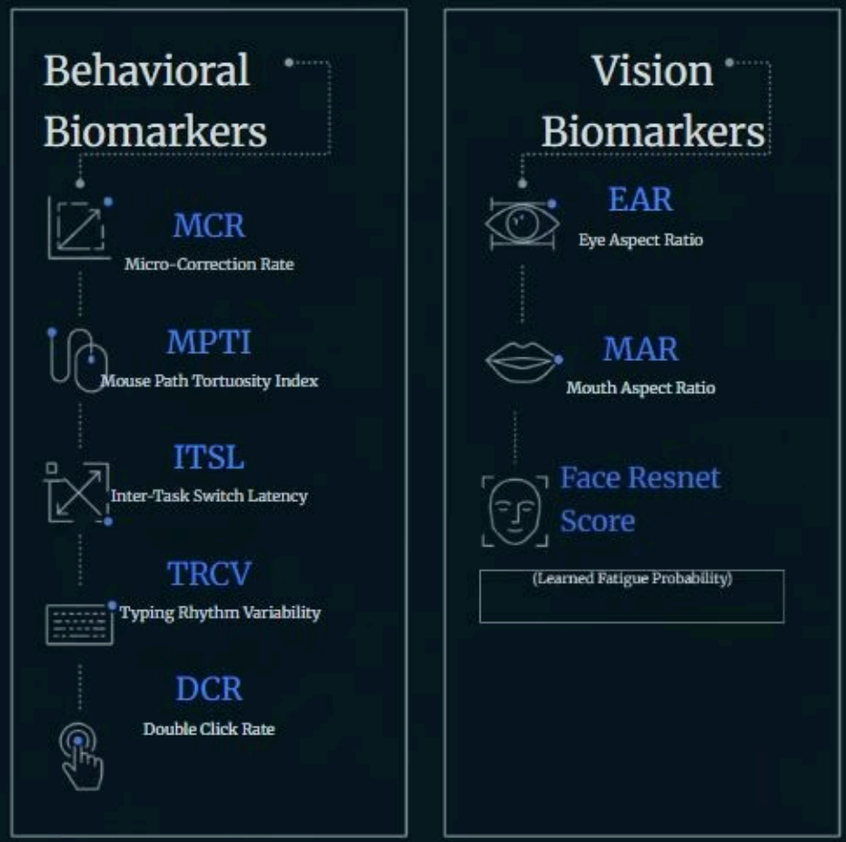
Final Output: Real-time fatigue confidence score + risk classification delivered instantly.

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Key Points

Behavioral & Vision Biomarkers

NeuroPulse AI tracks scientifically grounded biomarkers across two modalities — combining **how you look** with **how you behave** for unprecedented accuracy.



Why This Combination Is Unique

Most fatigue systems rely on a single signal. NeuroPulse AI fuses **physiological vision cues** with **behavioral interaction patterns** — creating a richer, more reliable fatigue profile than any single-modality approach.

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Core Innovation, Intervention & Privacy

Core Innovation

Two independent AI models — Vision Resnet and Behavioral XGBoost — fused into a single fatigue intelligence platform with a unified risk score.

Smart Interventions

Mild fatigue → break reminder · **High fatigue** → immediate alert · **Sustained overload** → wellness recommendation

Privacy-First Design

Local webcam processing only · No raw keystroke storage · No cloud dependency · Only derived metrics are analyzed

Mental Wellness Impact

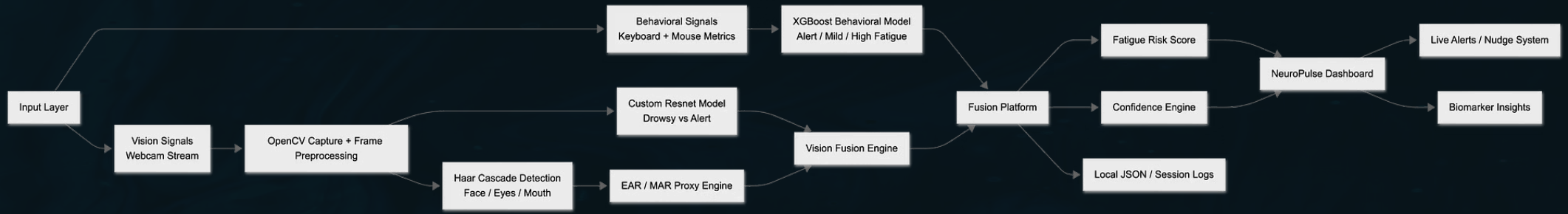
Proactively prevents burnout, overwork, and cognitive exhaustion before they become chronic problems.



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Complete AI Pipeline Architecture

NeuroPulse AI's end-to-end architecture processes vision and behavioral signals and fuses two independent AI models into one unified platform



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Technical Specification

Vision Stack

- OpenCV — frame capture & preprocessing
- OpenCV Haar Cascade — facial region detection
- PyTorch Resnet — drowsiness classification
- ONNX Runtime — optimized inference

Behavioral Stack

- Python — core pipeline logic
- XGBoost — gradient boosting classifier
- Scikit-learn — model evaluation & CV
- Streamlit — live dashboard UI

Dataset Strategy

Vision: Curated drowsy / non-drowsy facial dataset for binary classification training.

Behavioral: Hybrid dataset combining real typing benchmark data with simulated fatigue patterns — ensuring robust generalization.

5,546 samples in the final behavioral dataset, covering diverse user profiles and fatigue states.

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Model Performance Results

97.03%

Behavioral XGBoost Accuracy

Top-performing model on interaction-based fatigue signals

96.84%

5-Fold Cross Validation in XGBoost

Consistent generalization across all data splits

86.84%

Vision Resnet Accuracy

Binary drowsy vs. alert classification on facial imagery

Top Predictors Learned by the Model

1. Idle Time
2. Click Hesitation
3. Typing Variability
4. Eye Closure
5. Mouth Opening

Random Forest vs XGBoost Comparison

Metric	RF	XGBoost
Accuracy	96.31%	97.03%
CV Validation	96.52%	96.84%
Class Recall	strong	slightly better
Presentation Value	good	excellent

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Live Dashboard Demo

Behavioral Dashboard

- Focused Coder & High Fatigue mode views
- Manual input sliders for scenario testing
- Confidence bars & live biomarker display

Vision Mode

- Live webcam feed with real-time overlay
- EAR / MAR tracking indicators
- Continuous fatigue score + Alert/Drowsy detection

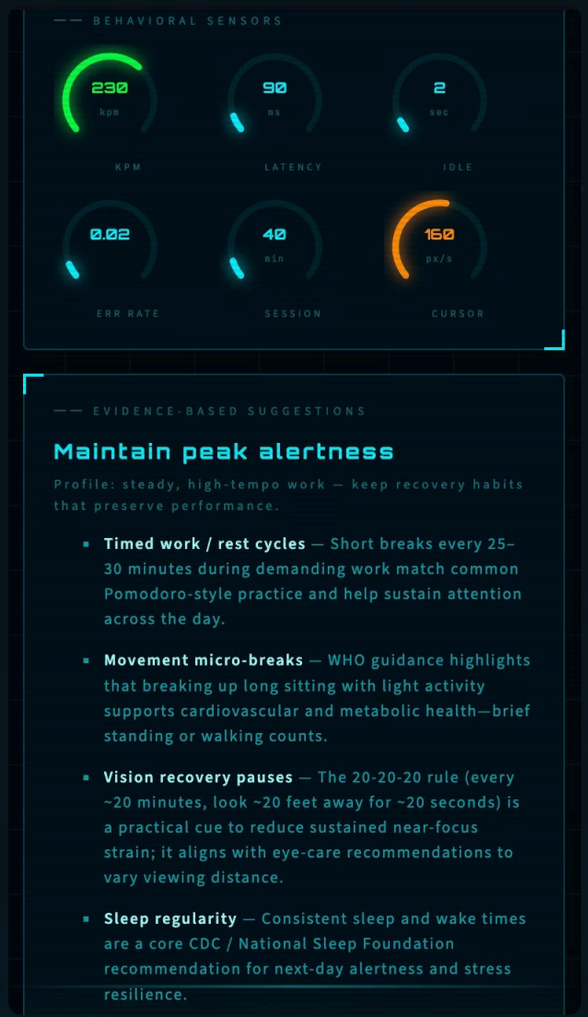
Explainable AI

- Transparent feature importance display
- Real-time reasoning behind each prediction
- Trustworthy, interpretable outputs

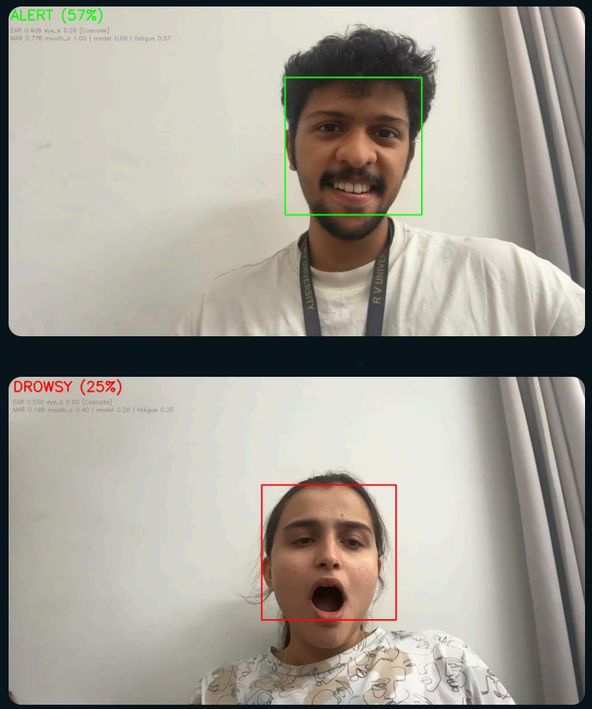
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Streamlit Dashboard





Behavioral Mode



Vision mode

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Future Scope & Conclusion

What's Next for NeuroPulse AI

- Unified multimodal fusion score
- Make it "edge-device ready" using TinyML technique.
- Smart adaptive nudges & wellness coaching
- Browser & IDE plugin integration
- Workplace wellness analytics platform
- Mobile app deployment

Conclusion

NeuroPulse AI combines **human behavior** and **human physiology** to detect fatigue before mistakes happen — a scalable, privacy-first AI system for productivity, wellness, and safety.

> *The future of AI is not just automation — it is protecting human performance.*

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